Project-1

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II

Note: Probably a much better way to do this so if you can clean it up feel free to change the code

```
## [1] "length"

## [1] 85

## [1] 9.595765

## [1] 2.029985

## [1] 6.7

## [1] 19.56

## [1] "infection"

## [1] 85

## [1] 4.388235

## [1] 1.35887

## [1] 1.3

## [1] 7.8

## [1] "census"

## [1] 85

## [1] 178

## [1] 147.1984

## [1] 37
```

```
## [1] 791

## [1] "xray"

## [1] 85

## [1] 81.02118

## [1] 19.15111

## [1] 39.6

## [1] 133.5

## [1] "facility"

## [1] 85

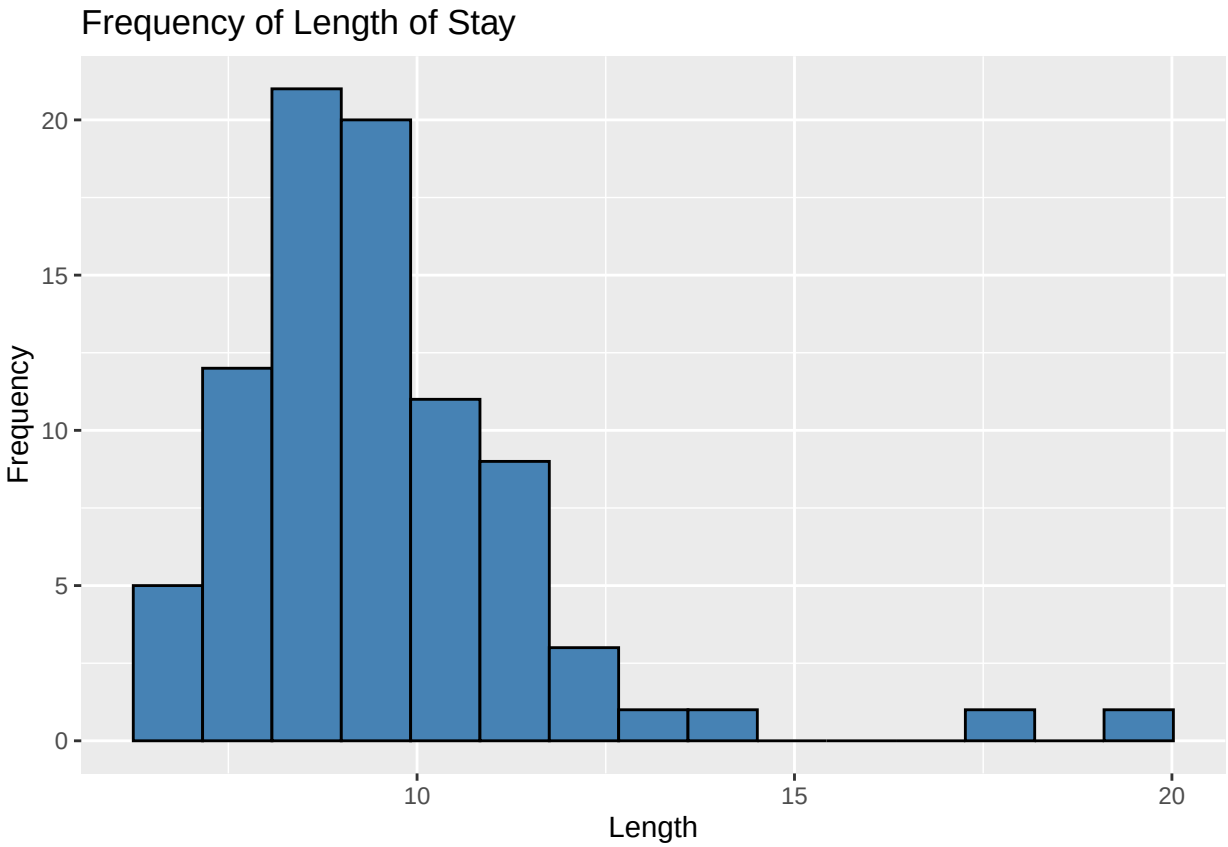
## [1] 41.94824

## [1] 15.04786

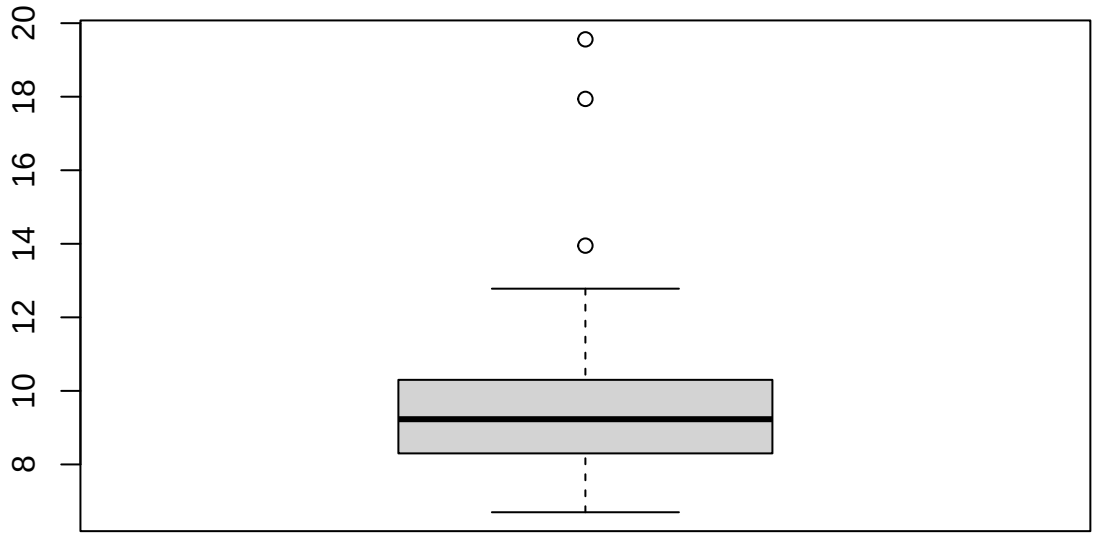
## [1] 5.7

## [1] 77.1
```

Distribution of length



Outliers of hospital length




```
## [1]  47 104 112
```

III

Intercept and Slope of Models

```
## (Intercept)
##    6.363955
```

```
## infection
## 0.7364713
```

```
## (Intercept)
##    8.496366
```

```
##      census
## 0.006176395
```

```
## (Intercept)
##    6.681713
```

```
##       xray
## 0.03596654
```

```
## (Intercept)
##    7.917803
```

```
##   facility
## 0.04000077
```

MSE and R^2

```
## [1] 3.082604
```

```
## [1] 3.255519
```

```
## [1] 3.603496
```

```
## [1] 3.714306
```

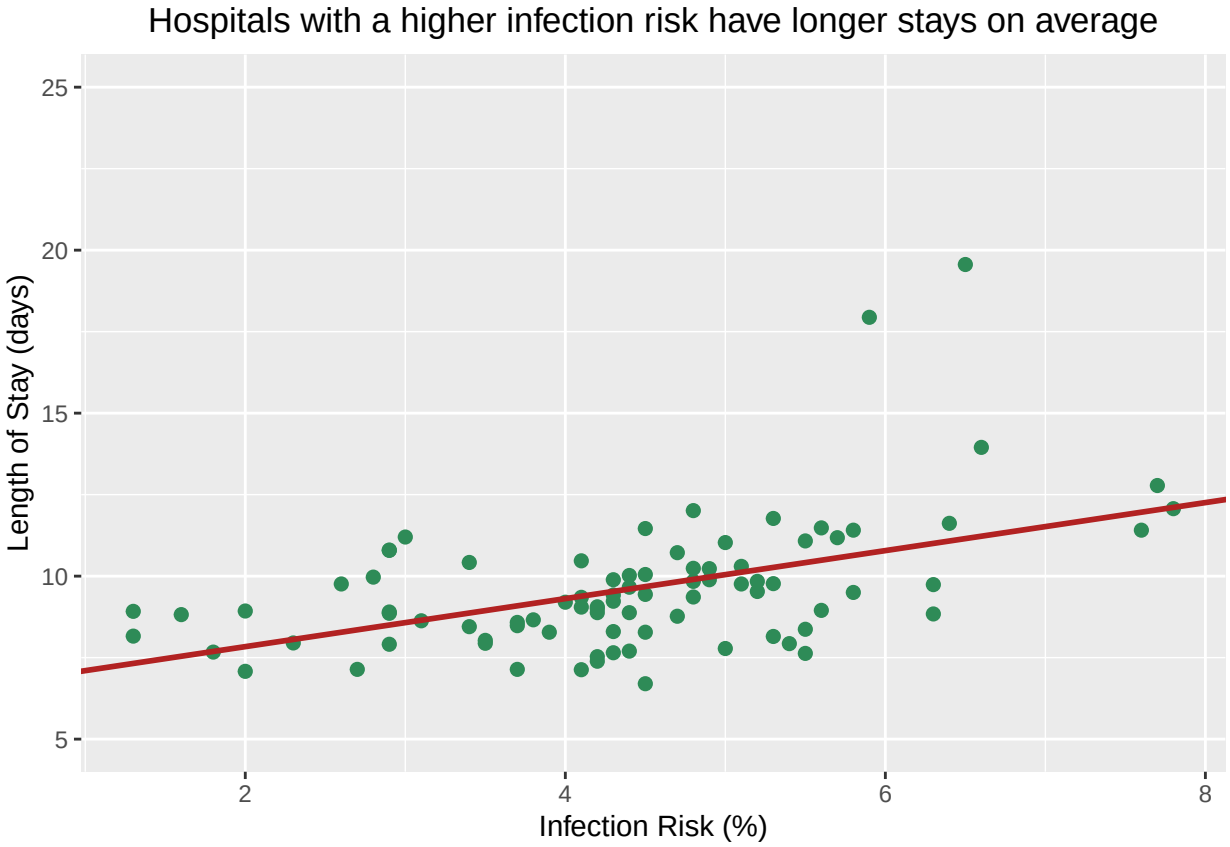
```
## [1] 0.2430421
```

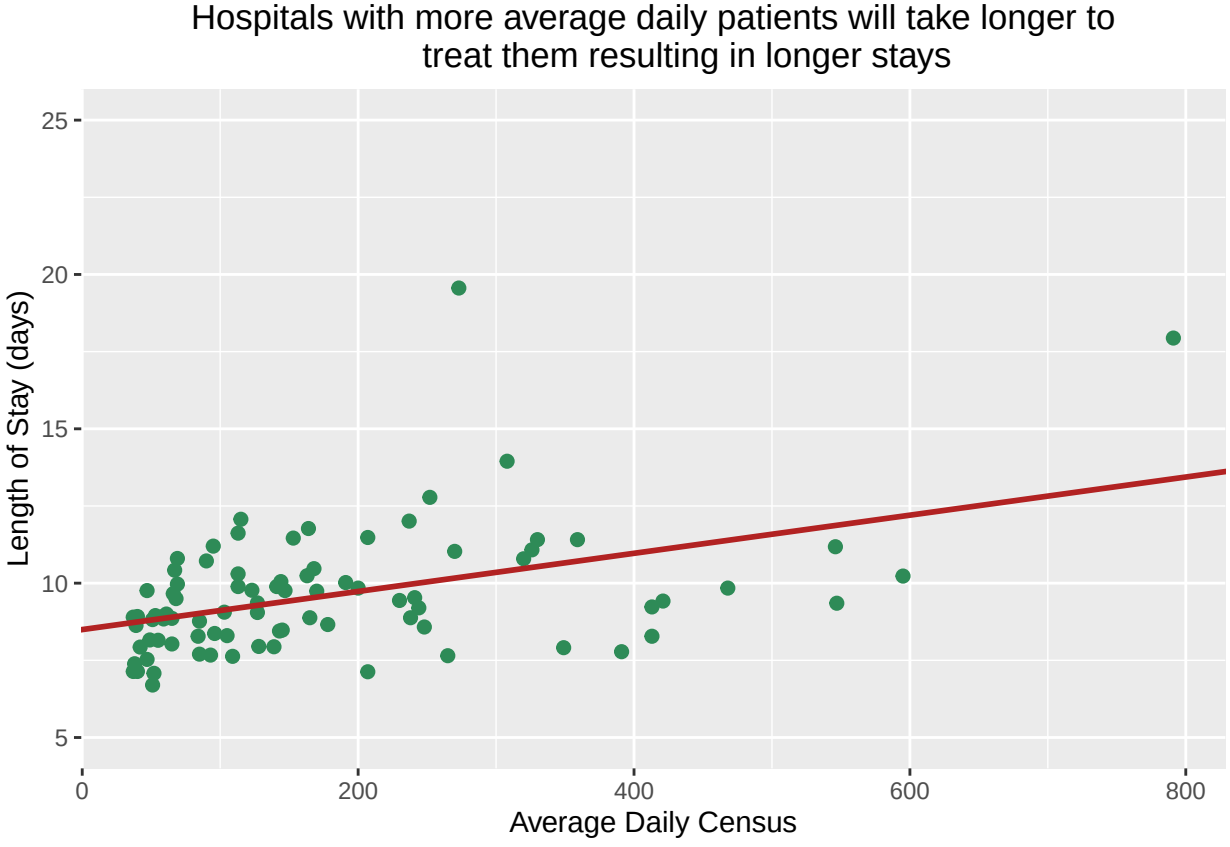
```
## [1] 0.2005815
```

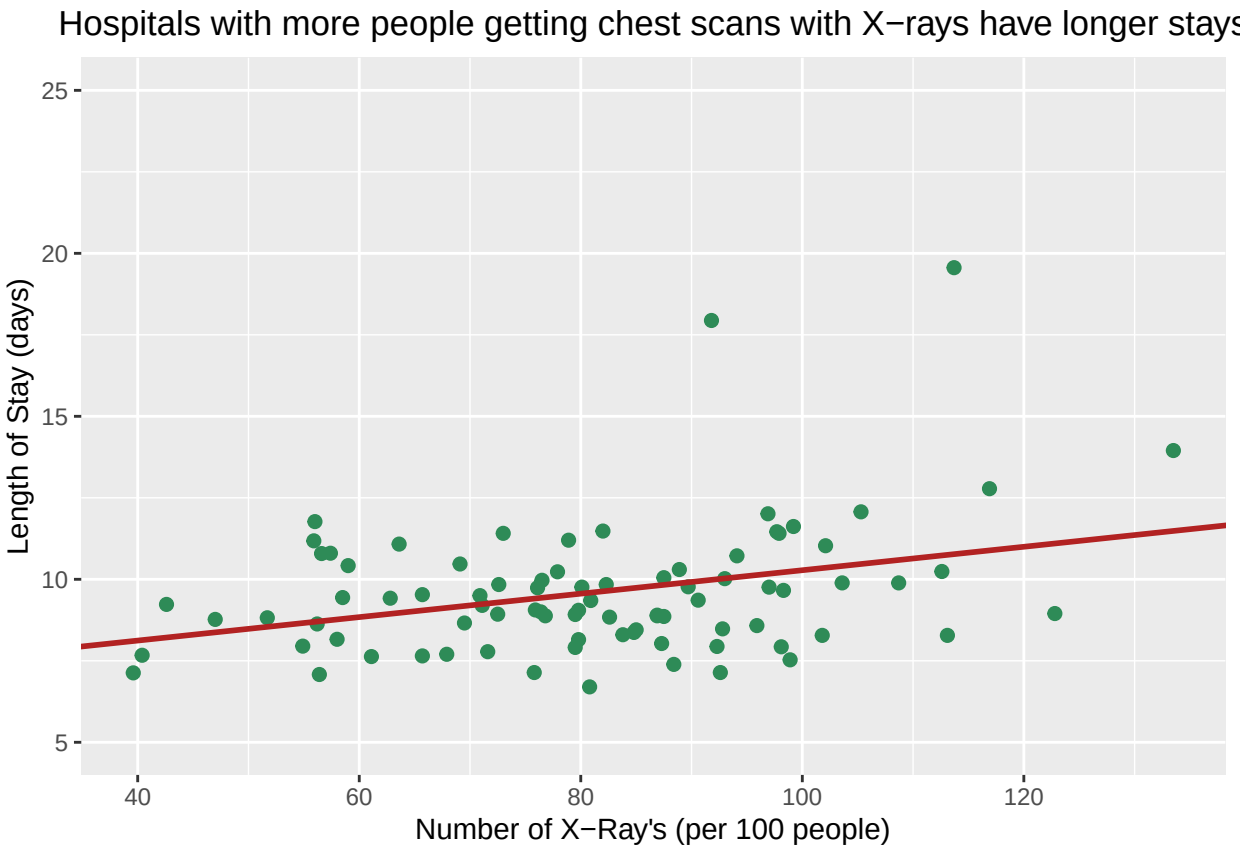
```
## [1] 0.1151329
```

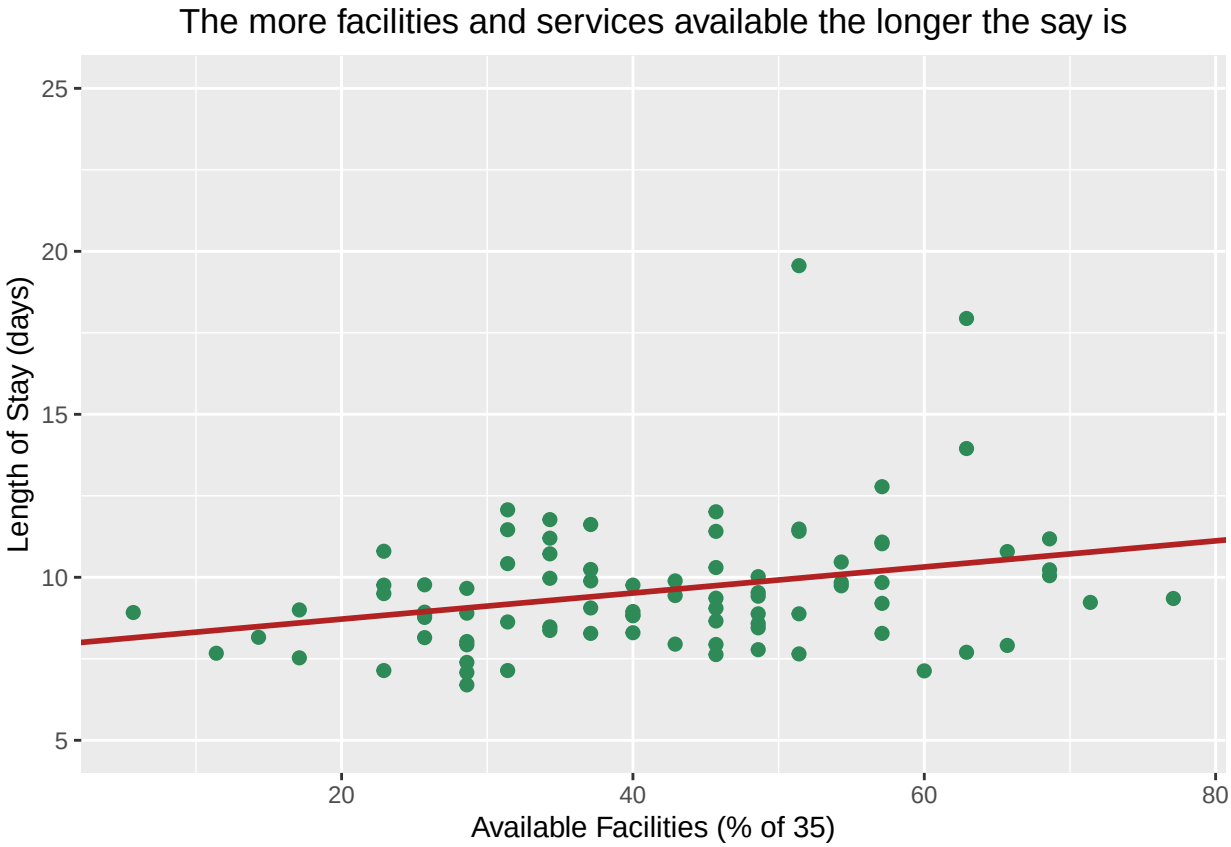
```
## [1] 0.08792258
```

Plots with regression line






Hospitals with more people getting chest scans with X−rays have longer stays
Length of Stay (days)
25
20
15
10
5
40
60
80
100
120
Number of X−Ray's (per 100 people)



Sorting by R^2

```
## [1] 0.2430421
```

```
## [1] 0.2005815
```

```
## [1] 0.1151329
```

```
## [1] 0.08792258
```

```
##          76          83          63          65           1          10
## -2.97807620 -2.78454751 -2.41090038 -2.26631186 -2.25348768 -2.16372456
##          94          71          45          57          21         111
## -2.11725325 -2.06713481 -2.04454751 -1.94889916 -1.92713481 -1.90442907
##          58           4          17          35          12         103
## -1.88078194 -1.53819464 -1.39807620 -1.26372456 -1.23078194 -1.21242785
##         110          84          82          26          70          72
## -1.13548890 -1.05537047 -1.00160490 -0.95619342 -0.91160490 -0.75689794
##          79          73          41          87          50          53
## -0.72442907 -0.66360612 -0.60889916 -0.58972211 -0.57713481 -0.55113726
##          89          68          97          95          15          75
## -0.53901760 -0.50889916 -0.50254629 -0.49725325 -0.45713481 -0.41795777
##          19           6          32          86          52         105
## -0.39713481 -0.35995899 -0.35360612 -0.33348768 -0.30078194 -0.23807620
```

```
##          66          25          99          30          24          54
## -0.11078194 -0.10984055 -0.10783933 -0.08266473 -0.05901760 -0.03843152
##          20          49          55          64          80          78
## -0.03348768 -0.01960368 -0.01701638  0.05557093  0.18004101  0.25733527
##          22         102          91          74          77          67
##  0.34098240  0.35921806  0.36027789  0.37192380  0.40027789  0.41557093
##          18          62          16          13          90          40
##  0.54262831  0.61815823  0.66545249  0.74521561  0.77451110  0.83863198
##          42          31          51          39          92           2
##  0.89462953  0.98368814  0.99180536  1.08651232  1.09310206  1.27769058
##         101          33          37          61          93          60
##  1.48121928  1.50274675  1.54392502  1.55204223  1.59863198  1.78192380
##          98          81         106          43         104         112
##  2.11098240  2.29027789  2.30027789  2.62663075  2.72533405  7.23086397
##          47
##  8.40898118
```

```
##          65          26          87           1          20          58
## -3.131336901 -2.767217598 -2.741928299 -2.644880169 -2.524854566 -2.483111096
##          76          78          52          72          66         103
## -2.111362505 -1.941321539 -1.817217598 -1.737538900 -1.676628760 -1.603422157
##          57          24          83          68          82          49
## -1.584892971 -1.546919338 -1.539593431 -1.448112376 -1.414885290 -1.400771107
##          71          99         111          21          50          97
## -1.341069366 -1.336944942 -1.321359945 -1.256656924 -1.086348423 -0.935764706
##          75          41          70          12          63          25
## -0.929590871 -0.911943662 -0.867832039 -0.844887850 -0.825774947 -0.803406795
##          17          45          62          94          40          79
## -0.735183549 -0.719300293 -0.688678170 -0.686068086 -0.639009714 -0.635471567
##         105          73          84          86          55          19
## -0.476937261 -0.454877609 -0.251359945 -0.230768546 -0.107245761 -0.072535060
##          91          10           2          92          89          32
## -0.037832039 -0.020773667  0.008637495  0.069226333  0.079231454  0.108354598
##           4          15          77          93          35          81
##  0.126284705  0.126873542  0.175107029  0.176577843  0.193646456  0.317187164
##          67           6          95         102          16         110
##  0.343942155  0.355703548  0.513937035  0.522761920  0.570128792  0.583638775
##          74          30          90          22          64          31
##  0.664232734  0.695700988  0.696307748  0.736881223  0.755991566  0.866006928
##          53          39         101          37          80          61
##  0.875423211  0.935999247  0.973343076  1.047462380  1.105700988  1.509815171
##          42          51         106          60          98          43
##  1.667758079  1.705119831  1.877462380  2.018645176  2.049827972  2.116876103
##          33          18          13          54         104         112
##  2.260704828  2.425700988  2.727182043  2.863348197  3.551303907  4.558104985
##          47
##  9.377477742
```

```
## [1]  47 112  76  83
```

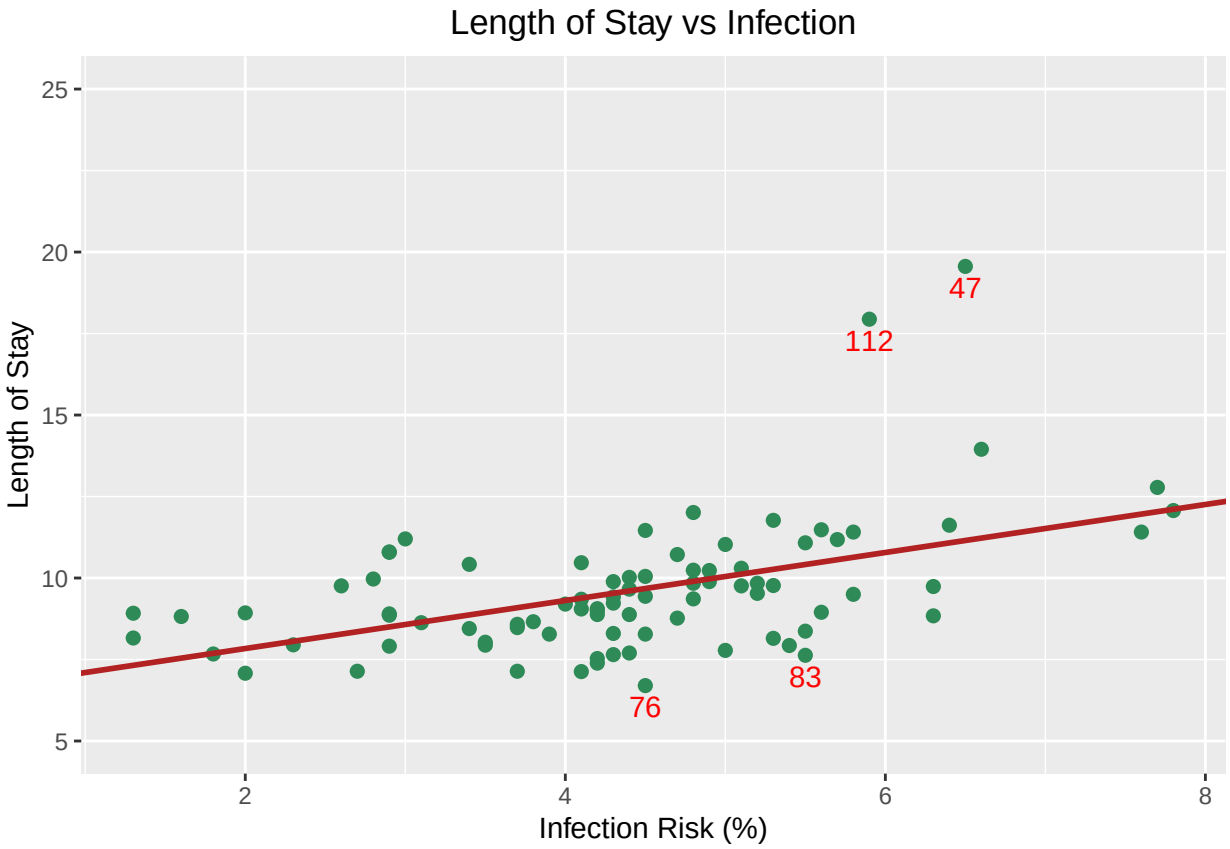
```
## [1] 47
```

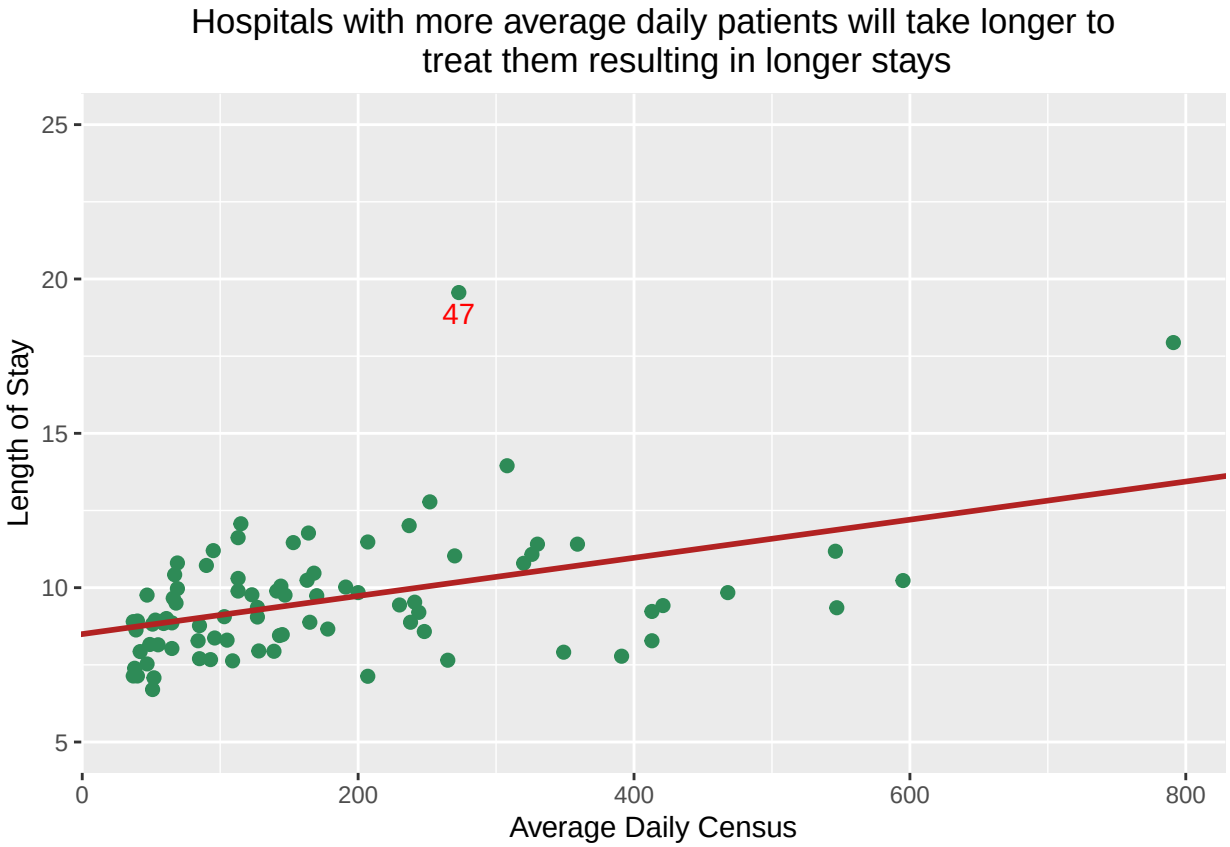
```
## [1] 0.2430421
```

```
## [1] 0.2833074
```

```
## [1] 0.2005815
```

```
## [1] 0.2376894
```





V

```
##             2.5 %   97.5 %
## infection 0.45272 1.020223
```

```
## [1] 5.162309
```

```
## [1] 1.64072e-06
```

```
## [1] 83
```

```
##        fit      lwr     upr
## 1 9.604429 9.221108 9.98775
```

```
##        fit      lwr      upr
## 1 9.604429 6.049792 13.15907
```

```
## [1] 1.98896
```

```
## [1] 2.492592
```

Table 1: Pointwise 95% CI and Working-Hotelling band

Xh	Yhat	CI_lower	CI_upper	WH_lower	WH_upper
2.5	8.21	7.55	8.86	7.38	9.03
4.5	9.68	9.29	10.06	9.20	10.16
7.0	11.52	10.68	12.35	10.47	12.56